

Primitives for Smoothing Mobile Robot Trajectories *

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Abstract: *Clothoids are very useful for smoothing the motion of a mobile robot moving along a trajectory. This paper addresses the problem of smoothing the mobile robot motions when cusps, i.e., changes of motion direction along the trajectory, are imposed. We pinpoint some special curves (that we call "anticlothoids") and we discuss how they can be used together with clothoids in order to smooth a predefined trajectory.*

1 Introduction

Clothoids (or Cornu spirals) are known as very useful curves to smooth trajectories. Their equation is $\kappa = k_c s + \kappa_0$ where κ is the curvature, s the arc length, κ_0 the initial curvature and k_c a constant characterizing the shape of the clothoid. Clothoids allow to link curves of infinite radius of curvature (i.e., lines) and curves of finite radius of curvature, with a continuous change of the curvature. Clothoids have practical applications in railway and highway design. They have been introduced in Robotics ([5, 2, 9]) for smoothing the motions of a mobile robot moving "forward" on a broken line (i.e., without a change of orientation along the trajectory).

This paper pinpoints the *involute* of circles¹. The natural equation of an involute of a circle is $\rho^2 = 2k_a s$, ρ being the radius of curvature and k_a the radius of the circle. Another general form of this equation is $\rho = k_a(\theta - \theta_0) + \rho_0$ where θ is the direction of the tangent, θ_0 the initial direction of the curve² and ρ_0 the initial radius of curvature. k_a is the characteristic parameter of the involute. Such curves allow to link "curves" of infinite curvature (i.e., curves reduced to a point) and curves of finite curvature, with a continuous change of the curvature. This property of the circle involutes leads us to call them **anticlothoids** in the context of this paper. Like clothoids, anticlothoids are the time-optimal trajectories of a two driving wheels mobile robot [6]. Both types of curves are dual from the point of view of control. They are produced by applying respectively a same constant acceleration on both wheels (anticlothoids) or constant

and opposite accelerations (clothoids). The purpose of this paper is to show how to use anticlothoids in order to smooth the motions of a mobile robot when cusps are imposed (i.e., when the robot has to change the direction of its motion).

We shall first overview some known results on trajectory smoothing, mainly using clothoids (§2). Then we introduce both types of curves from a control theory viewpoint, and we show how a mobile robot can execute motions supported by them (§3).

Geometric properties are then proven (§4). Two connected oriented straight line segments being given as a reference trajectory, we show how to plan a motion such that:

1. the velocities of the driving wheels are continuous and never simultaneously null (i.e the motion is smooth),
2. the trajectory never lies farther than some fixed threshold from the segments (this condition is required for collision avoidance).
3. the produced trajectory respects the imposed directions of motion.

The trajectories will consist of sequences of clothoids and anticlothoids.

In a last section (§5), we shall conclude by sketching a method for smoothing a mobile robot's trajectory consisting of a polygonal line, while accounting for non-collision and direction changes.

2 Related work

Clothoids are extensively used for the design of highways [4] in order to smoothly join straight lines with circular portions. They are also used as splines in Computer Aided Design (CAD) [12].

They were more recently introduced in Robotics. Indeed, most trajectory planners produce trajectories consisting of straight lines and turns that force the robot to stop, because of the discontinuity in the angular speed when it has to change direction. In order to eliminate these stops, smoothing of the reference trajectory to produce a swift motion has been a long-time objective, for which the use of clothoids is very popular.

The smoothing problem was also addressed for producing directly a trajectory that joins a sequence of robot configurations $(x, y, \theta)_i$ smoothly, i.e., without stopping.

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¹The involute of a circle is the curve described by the end of a thread as it is unwound from a stationary spool [7]

²As s is a measure of the length of the curve, $(\theta - \theta_0)$ measures the angular variation of the curve.

This problem has been addressed by Iijima, Kanayama and Yuta [5] where clothoid curves were exhibited as transition trajectories. The use of clothoids was also studied and implemented on the mobile robot Hilare [2, 10] for smoothing a path consisting of straight lines and turns.

In [9], Kanayama and Miyake join initial and final configurations by a broken line that is used as an entry to a smoothing algorithm which produces sequences of clothoid arcs and straight lines.

Shin and Singh [15] address a similar problem with the additional constraint of a lower bounded gyration radius. They produce a smooth path composed of clothoid arcs only.

In [8], Kanayama and Hartman propose the use of cubic spirals for obtaining smooth trajectories linking a sequence of configurations two by two. They propose a criterion for smoothness (a quadratic function of the curvature or of its derivative). The use of cubic spirals produces trajectories of a larger curvature radius than in the case of clothoids, thus producing a "smoother" motion.

Delingette *et al.* [3] generalize the smoothing problem to take into account given end conditions as well as a limited gyration radius by adding control points to the trajectory. This is much similar to the approaches in CAD. They introduced "intrinsic splines", curves whose curvature is a polynomial function of the arc length. These curves are generalizations of the clothoids and cubic spirals.

The involutes of circles were never exploited for smoothing purposes, to our knowledge. They actually correspond to the smoothing of *cusps* (seldom encountered in highways...). In robotics however, the generation of collision-free trajectories sometimes imposes such constraints. As we shall see, the involute of a circle then becomes a very useful curve.

3 Clothoids and anticlothoids

3.1 Time-optimal trajectories for a two driving wheel mobile robot

Let us consider mobile robots whose locomotion system consists of two parallel independant driving wheels (e.g., the three mobile robots of the Hilare family developed at Laas [1]). The length of the axis supporting the wheels is noted d . The reference point of the robot is the midpoint of the axis. Its coordinates are denoted by (x, y) . The direction of the vehicle is noted θ . Therefore, a configuration of the robot is a triple $(x, y, \theta) \in (\mathbb{R}^2 \times S^1)$. We denote by v_r and v_l the linear velocities of the right and left wheels respectively (see Figure 1). The robot can then be modeled by the following control system:

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{v}_r \\ \dot{v}_l \end{pmatrix} = \begin{pmatrix} \frac{1}{2}(v_r + v_l) \cos \theta \\ \frac{1}{2}(v_r + v_l) \sin \theta \\ \frac{1}{d}(v_r - v_l) \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} u_r + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} u_l$$

Moreover, the accelerations u_r and u_l of the wheels (i.e. the inputs of the system) are assumed to be bounded : $|u_r| \leq a$ and $|u_l| \leq a$.

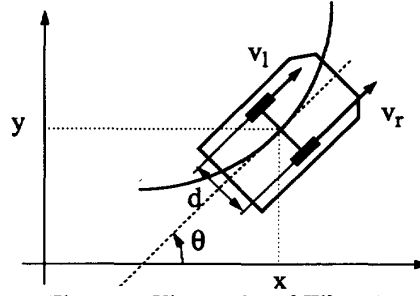


Figure 1: Kinematics of Hilare 2.

By applying the Minimum Principle (from optimal control theory [13]), one can prove that the time-optimal controls of the above system verify $|u_r| = |u_l| = a$ [6]. Thus the time-optimal trajectories are supported by two types of curves corresponding respectively to the cases $u_r = u_l$ and $u_r = -u_l$.

Remarks:

1. This result does not characterize the shape of the time-optimal trajectories linking two given configurations. It just gives necessary conditions for a trajectory to be time optimal (i.e., it shows how the optimal trajectory looks like *locally*). At this moment, providing sufficient conditions of optimality is still an open problem (see [14] for a numerical approach to the problem).
2. We can relate u_r and u_l to the linear and angular accelerations $\dot{v}$ and $\dot{\omega}$ by the following equation:

$$\begin{pmatrix} \dot{v} \\ \dot{\omega} \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/d & -1/d \end{pmatrix} \begin{pmatrix} u_r \\ u_l \end{pmatrix}$$

Thus, the optimal trajectories presented here are not only valid for the two-driving wheels mobile robots, but also for any system with linear and angular acceleration controls, without constraints on the curvature.

Now we show that the two types of curves have dual geometrical properties, as announced in the introduction.

Notations : A motion of the robot consists of two mappings u_r and u_l from a time interval $[0, 1]$ onto $\{-a, a\}$. The corresponding trajectory of the robot (i.e., the locus of the points (x, y) in the plane) is noted γ . γ is a mapping from $[0, 1]$ onto the plane $\mathbb{R}^2$. The arc length from $\gamma(0)$ to $\gamma(t)$ is noted $s(t)$. The curvature at a point $\gamma(t)$ is noted $\kappa(t)$, while the radius of curvature is $\rho(t) = 1/\kappa(t)$. The linear and angular velocities are respectively noted $v(t)$ and $\omega(t)$. The direction of the tangent to γ at some point $\gamma(t)$ (when it is defined) is the direction θ of the vehicle at this point, i.e., $\omega(t) = \dot{\theta}(t)$. Finally recall that $\rho(t) = v(t)/\omega(t)$.

All the initial values $v(0)$, $\omega(0)$, $\theta(0)$... are noted v_0 , ω_0 , θ_0 ... The final values are noted v_1 , ω_1 , θ_1 ...

3.2 Case $u_r = -u_l$: the clothoids

Let us consider two constant controls verifying $u_r(t) = -u_l(t)$. In this case $\dot{v}(t) = \frac{1}{2}(\dot{v}_r(t) + \dot{v}_l(t)) = 0$. $v(t)$ is constant and equal to v_0 . Then $s(t) = v_0 t$.

Now, $\dot{v}_r(t) = \pm a$ while $\dot{v}_l(t) = \mp a$. By integration, $v_r(t) = \pm at + v_{r0}$, $v_l(t) = \mp at + v_{l0}$ and $\omega(t) = \dot{\theta}(t) = \pm \frac{2a}{d}t + \omega_0$.

Therefore,

$$\kappa(t) = \frac{\pm \frac{2a}{d}t + \omega_0}{v_0} = \pm \frac{2a}{dv_0^2}s(t) + \frac{\omega_0}{v_0} \quad (1)$$

The curve is a clothoid whose characteristic constant k_c is $\frac{2a}{dv_0^2}$.

When $\omega_0 = \theta_0 = 0$, the coordinates of a point $\gamma(t)$ are given by the Fresnel integrals (see [7] for instance):

$$\begin{aligned} x(t) &= \text{sgn}(v_0) \sqrt{\frac{\pi}{k_c}} \int_0^{\sqrt{\frac{2a}{d\pi}}t} \cos \frac{\pi}{2} u^2 du \\ &= \text{sgn}(v_0) \sqrt{\frac{\pi}{k_c}} CF\left(\sqrt{\frac{2a}{d\pi}}t\right) \end{aligned} \quad (2)$$

$$\begin{aligned} y(t) &= \text{sgn}(v_0 u_r) \sqrt{\frac{\pi}{k_c}} \int_0^{\sqrt{\frac{2a}{d\pi}}t} \sin \frac{\pi}{2} u^2 du \\ &= \text{sgn}(v_0 u_r) \sqrt{\frac{\pi}{k_c}} SF\left(\sqrt{\frac{2a}{d\pi}}t\right) \end{aligned} \quad (3)$$

The tangent direction at this point is :

$$\theta(t) = \text{sgn}(u_r) \frac{a}{d} t^2 \quad (4)$$

Figure 2 shows a clothoid curve for $u_r > 0$. For an initial configuration $(0, 0, 0)$ the upper part is described for $v_0 > 0$, and the lower part for $v_0 < 0$. A curve symmetrical with respect to the Ox axis is obtained for $u_r < 0$. Computation of the coordinates when $\omega_0 \neq 0$ is easily done from the above system.

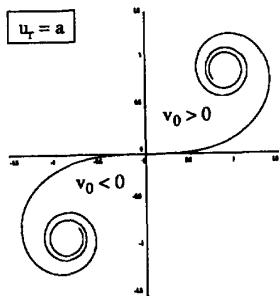


Figure 2: A clothoid for $u_r > 0$. Direction of motion along the curve is determined by the sign of v_0 .

Remarks:

1. In the computations above, we assume $v_0 \neq 0$. When $v_0 = 0$, direct computations show that γ corresponds to a rotation³. Rotations can then be viewed as degenerated clothoids with infinite characteristic constants (i.e., points).
2. The linear velocity v along a clothoid is constant.
3. Equation (1) shows that clothoids can link smoothly curves of zero curvature (lines) and curves with non zero curvature.

3.3 Case $u_r = u_l$: the anticlothoids

Now, let us consider two constant controls verifying $u_r(t) = u_l(t)$. In this case, we have $\dot{\omega}(t) = \frac{1}{d}(\dot{v}_r(t) - \dot{v}_l(t)) = 0$ and $\omega(t)$ is constant and equal to ω_0 . Thus, $\theta(t) = \omega_0 t + \theta_0$.

The acceleration $\dot{v}(t)$ equals $\text{sgn}(u_r)a$. Thus, $v(t) = \text{sgn}(u_r)at + v_0$. Therefore,

$$\rho(t) = \frac{v(t)}{\omega(t)} = \frac{\text{sgn}(u_r)at + v_0}{\omega_0} = \text{sgn}(u_r) \frac{a}{\omega_0^2} (\theta(t) - \theta_0) + \frac{v_0}{\omega_0} \quad (5)$$

The curve is an anticlothoid whose characteristic constant is $k_a = a/\omega_0^2$.

When $v_0 = \theta_0 = 0$, the coordinates of a point $\gamma(t)$ are given by the following parametric system :

$$x(t) = \text{sgn}(u_r) k_a (\cos(\omega_0 t) + \omega_0 t \sin(\omega_0 t) - 1) \quad (6)$$

$$y(t) = \text{sgn}(u_r) k_a (\sin(\omega_0 t) - \omega_0 t \cos(\omega_0 t)) \quad (7)$$

The tangent direction at this point is :

$$\theta(t) = \omega_0 t \quad (8)$$

Figure 3 shows an anticlothoid curve for $u_r > 0$. For an initial configuration $(0, 0, 0)$, a positive ω_0 causes a counterclockwise motion, whereas a negative ω_0 causes a clockwise motion. A curve symmetrical with respect to the origin is obtained for $u_r < 0$. The computation of the coordinates for the general case ($v_0 \neq 0$, $\theta_0 \neq 0$) is given in the appendix.

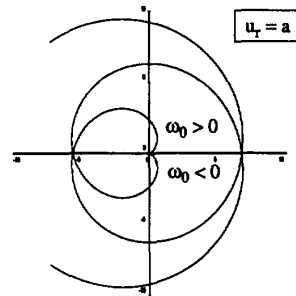


Figure 3: An anticlothoid for $u_r > 0$.

³In this paper, a rotation always designates a turn around the robot's own center (axis).

Remarks:

1. When $\omega_0 = 0$, γ is a line. Lines appear as degenerated anticloths with infinite characteristic constants.
2. The angular velocity ω along an anticlothoid is constant.
3. Equation (5) shows that anticloths can link smoothly curves with zero radius of curvature to curves with a non zero curvature radius.
4. While anticloths and clothoids are dual curves from the control theory point of view, the first ones are analytic curves and the second ones are not.

3.4 How to link clothoids and anticloths?

Any motion starts with $v_0 = \omega_0 = 0$ and ends with $v_1 = \omega_1 = 0$. This means that any motion provided by optimal controls starts and ends either with a rotation or a translation.

Moreover, since the angular velocity is zero along a line, the trajectory portion after or before a line, if any, is necessarily a clothoid. In the same way, the portion after or before a rotation is necessarily an anticlothoid.

Figure 4 shows all the possible combinations linking clothoids and anticloths. The type of trajectories are the nodes of the graph, while the conditions of switching are the arcs.

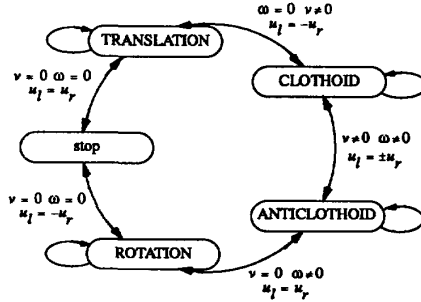


Figure 4: Possible combinations of clothoids and anticloths.

Now, let us consider a trajectory γ consisting of an arc of clothoid followed by an arc of anticlothoid, such that $\omega_0 = 0$ and $v_1 = 0$. Such a trajectory links a translation and a rotation (we fix $\theta_0 = 0$). Let $\tau \in [0, 1]$ such that $\gamma(\tau)$ is the meeting point of the two arcs. We note $\theta(\tau), \omega(\tau), v(\tau)$ respectively θ_τ, ω_τ , and v_τ .

With $u_r > 0$, we have from equation (4), $\theta_\tau = \frac{a}{d}\tau^2$. From equation (8), $\theta_1 = (1 - \tau)\omega_\tau + \theta_\tau$. Because the linear velocity is constant along a clothoid, $v_\tau = v_0$. Now, $v_1 = 0$, thus $v_0 = (1 - \tau)a$. Because the angular velocity is constant along an anticlothoid and $\omega_0 = 0$, we have $\omega_1 = \omega_\tau = \frac{2a}{d}\tau$.

The characteristic constants of the clothoid and the anticlothoid are respectively $k_c = \frac{2a}{d\omega_\tau^2}$ and $k_a = \frac{a}{\omega_\tau^2}$.

Thus:

$$\theta_\tau = \frac{d\omega_\tau^2}{4a} = \frac{d}{4k_a} \quad (9)$$

$$\theta_1 = \theta_\tau + \frac{\omega_1 v_0}{a} = \frac{d}{4k_a} + \sqrt{\frac{2}{dk_c k_a}} \quad (10)$$

Equation (10) gives the angular variation along γ as a function of the characteristic parameters k_a and k_c . It is used for proving the correctness of the smoothing primitives below.

4 Four Primitives for Smoothing Paths and their Geometric Properties

Consider a collision-free trajectory consisting of a sequence of straight-line segments. All the segments are directed. The direction of the segments gives the direction of the movement (forward or backward) on these segments. The endpoints of the segments are also directed according to the direction of the rotations at these points. Such a trajectory can be *a priori* given to the robot by a user or provided by some motion planner (e.g. [11]).

Linking a translation and a rotation forces the robot to stop. At the stop point, *both* linear and angular velocities v and ω are zero. The trajectory appears as a sequence such as (Stop-TRANSLATION-Stop-ROTATION-Stop-TRANSLATION-...-Stop)⁴. The goal of the smoothing is to remove the stop points, while maintaining the continuity of the velocities v and ω , and controlling the deviation of the trajectory after smoothing from the initial one. At any point of the resulting trajectory, either ω or v are non zero.

4.1 The canonical motion [T-s-R-s-T]

To this end, we introduce canonical primitives for smoothing sub-sequences of trajectory such as [TRANSLATION-Stop-ROTATION-Stop-TRANSLATION] (or [T-s-R-s-T]). We want to produce a smooth trajectory symmetrical with respect to the bisector of the lines supporting consecutive translations separated by a rotation.

Notations : We note S_1 the first straight line segment and S_2 the second one (see figure 5). P is the intersection point of S_1 and S_2 . α is the measure of the angle of the two lines supporting S_1 and S_2 ($0 < \alpha < \pi$; the case $\alpha < 0$ can be deduced by symmetry). Moreover we assume that the direction of motion on S_1 is positive (i.e., the robot moves forward). The case of a backward motion can also be deduced by symmetry. We note σ the direction of motion on S_2 : $\sigma = 1$ if the robot moves forward and $\sigma = -1$ otherwise. Finally, δ is the direction of rotation at P : $\delta = 1$ if the robot rotates counterclockwise and $\delta = -1$ otherwise.

The inputs of the problem are an elementary sequence [T-s-R-s-T] and a safety parameter defined for collision avoidance, i.e.:

- two straight line segments S_1 and S_2 , their length e , intersection point P and angle α .

⁴or: (Stop-ROTATION-Stop-TRANSLATION-...-Stop)

- σ the imposed direction of motion on S_2 .
- δ the imposed direction of rotation at P .
- a positive number ϵ measuring the maximum deviation between the reference trajectory (S_1, S_2) , and the resulting smoothed trajectory.

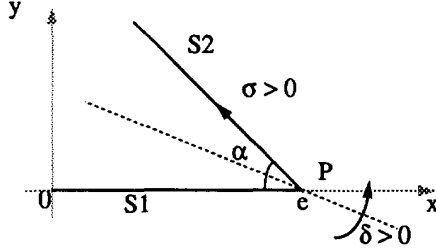


Figure 5: Example of inputs of the problem.

The output is a trajectory γ such that:

- the initial and final configurations of γ are the initial and final configurations of the input trajectory.
- γ is symmetric with respect to the bisector (S_1, S_2) .
- the velocities of the wheels of the mobile robot moving along γ are continuous and never simultaneously null.
- the variation of the vehicle direction along γ is *exactly* the variation of the vehicle direction during the rotation, with the sign of δ .
- the distance of any point on γ to the reference trajectory (S_1, S_2) is never greater than ϵ .

According to the directions of motion on S_2 ($\sigma = \pm 1$) and during the rotation ($\delta = \pm 1$), we have to consider four cases.

In all cases, the trajectories to compute must start and end at configurations with zero curvature. Therefore, the initial and final portions of the trajectories are arcs of clothoids.

When $\sigma = -1$, the linear velocity of the robot has to be zero at some point because of its direction change. Thus, the trajectories to compute will contain arcs of anticlothoids.

4.2 $\sigma = 1$ and $\delta = 1$: the turn

This is a classical case. The smoothing primitive consists of two arcs of clothoid that meet at a configuration c such that:

$$\begin{cases} x_c = e - \epsilon \cot \frac{\alpha}{2} \\ y_c = \epsilon \\ \theta_c = \frac{\pi - \alpha}{2} \end{cases}$$

We want to compute an arc of clothoid linking the x -axis to c . From equation (4) we can substitute t by

$\sqrt{\frac{d\theta_c}{dt}}$ in equations (2) and (3) :

$$\begin{aligned} x_c &= \sqrt{\frac{\pi}{k_c}} CF\left(\sqrt{\frac{2\theta_c}{\pi}}\right) + x_0 \\ y_c &= \sqrt{\frac{\pi}{k_c}} SF\left(\sqrt{\frac{2\theta_c}{\pi}}\right) \end{aligned}$$

$(x_0, 0, 0)$ is the initial configuration of the clothoid. Now, we have to compute k_c and x_0 :

$$\begin{aligned} x_0 &= e - \epsilon \left(\cot \frac{\alpha}{2} + CF\left(\sqrt{\frac{\pi - \alpha}{\pi}}\right) / SF\left(\sqrt{\frac{\pi - \alpha}{\pi}}\right) \right) \\ k_c &= \frac{\pi}{\epsilon^2} SF\left(\sqrt{\frac{\pi - \alpha}{\pi}}\right) \end{aligned}$$

The first clothoid arc is then computed. The second one is deduced by symmetry. Figure 6 shows the resulting smoothed trajectory.

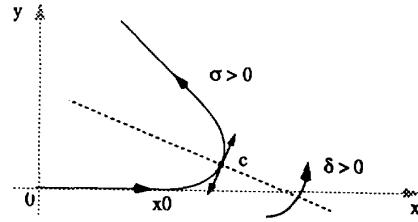


Figure 6: The turn.

4.3 $\sigma = 1$ and $\delta = -1$: the loop

This (less classical) case is similarly processed by reasoning from the intermediate configuration c . the result is a *loop* composed of two clothoid arcs (see figure 7) :

$$\begin{cases} x_c = e + \epsilon \cos \frac{\alpha}{2} \\ y_c = -\epsilon \sin \frac{\alpha}{2} \\ \theta_c = -\frac{\pi + \alpha}{2} \end{cases}$$

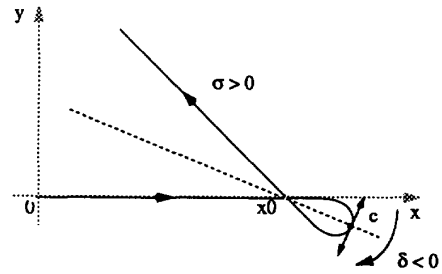


Figure 7: The loop.

4.4 $\sigma = -1$ and $\delta = 1$: the internal cusp

The case of cusps is more complicated. The primitive of smoothing consists of two arcs of clothoid linked

by two arcs of anticloithoid, symmetrical with respect to the bisector of S_1 and S_2 .

Consider the configuration c lying on the bisector and verifying :

$$\begin{cases} x_c = e - \epsilon \cot \frac{\alpha}{2} \\ y_c = \epsilon \\ \theta_c = \frac{2\pi - \alpha}{2} \end{cases}$$

We build a sequence consisting of an arc of a clothoid and an arc of an anticloithoid starting from the x -axis to c .

Let $\tilde{c}$ be the configuration at the switching between the clothoid and the anticloithoid. Being the meeting point of a clothoid and an anticloithoid, $\tilde{c}$ verifies equation (9) and:

$$\begin{aligned} x_{\tilde{c}} &= \sqrt{\frac{\pi}{k_c}} CF\left(\sqrt{\frac{2\theta_a}{\pi}}\right) + x_0 \\ y_{\tilde{c}} &= \sqrt{\frac{\pi}{k_c}} SF\left(\sqrt{\frac{2\theta_a}{\pi}}\right) \\ \text{with } \theta_{\tilde{c}} &= \frac{d}{4k_a} \end{aligned}$$

Now, c is the endpoint of an arc of an anticloithoid whose initial configuration is $\tilde{c}$. From the computations done in Section 2.4 (equation 10) and in the appendix (equations (11) and (12)) :

$$\begin{cases} x_c = k_a(\cos \theta_c - \cos \theta_{\tilde{c}} + (\theta_c - \theta_{\tilde{c}}) \sin \theta_{\tilde{c}}) + x_{\tilde{c}} \\ y_c = k_a(\sin \theta_c - \sin \theta_{\tilde{c}} - (\theta_c - \theta_{\tilde{c}}) \cos \theta_{\tilde{c}}) + y_{\tilde{c}} \\ \theta_c = \frac{d}{4k_a} + \sqrt{\frac{2}{dk_c k_a}} \end{cases}$$

Therefore the indeterminates x_0 , k_a and k_c verify the following system :

$$\begin{cases} e - \epsilon \cot \frac{\alpha}{2} = k_a(\cos \theta_c - \cos \frac{d}{4k_a} + \sqrt{\frac{2}{dk_c k_a}} \sin \frac{d}{4k_a}) \\ \quad + \sqrt{\frac{\pi}{k_c}} CF\left(\sqrt{\frac{d}{2k_a \pi}}\right) + x_0 \\ \epsilon = k_a(\sin \theta_c - \sin \frac{d}{4k_a} - \sqrt{\frac{2}{dk_c k_a}} \cos \frac{d}{4k_a}) \\ \quad + \sqrt{\frac{\pi}{k_c}} SF\left(\sqrt{\frac{d}{2k_a \pi}}\right) \\ \frac{2\pi - \alpha}{2} = \frac{d}{4k_a} + \sqrt{\frac{2}{dk_c k_a}} \end{cases}$$

From the third equation we can compute k_c as a function of k_a . By replacing k_c in the second equation, we obtain an equation with the only indeterminate k_a . We solve this new equation numerically. Finally x_0 is given by the first equation.

Figure 8 shows an example of trajectory composed of this primitive.

4.5 $\sigma = -1$ and $\delta = -1$: the external cusp

Here, the primitive is exactly the same as the previous one, except that we consider the intermediate following configuration c (Figure 9) :

$$\begin{cases} x_c = e + \epsilon \cos \frac{\alpha}{2} \\ y_c = -\epsilon \sin \frac{\alpha}{2} \\ \theta_c = -\frac{\alpha}{2} \end{cases}$$

The smoothed trajectory lies on the other side of the reference trajectory (S_1, S_2) (figure 9).

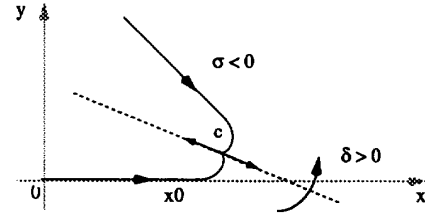


Figure 8: The internal cusp.

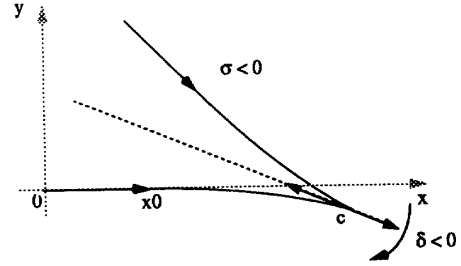


Figure 9: The external cusp.

4.6 Geometrical property for collision avoidance

All the above trajectories are clearly produced by smooth controls. According to the kinematic properties of the motions along clothoids and anticloithoids, the velocities of the wheels of the mobile robot are never zero simultaneously.

We have now to check that all the four trajectories above verify the geometric constraints imposed by the inputs.

Property : The four primitives described above are trajectories γ lying at a distance less than ϵ from S_1 and S_2 .

Proof : The proof comes from the choice of the intermediate configuration c lying on the bisector of segments S_1 and S_2 , and from the convexity of the arcs of clothoids and anticloithoids used to construct γ in each case.

Consider the “tube” consisting of all the points whose distance to S_1 and S_2 is less than ϵ . It is bounded by :

- two segments intersecting at point C_1 with coordinates $(e - \epsilon \cot \frac{\alpha}{2}, \epsilon)$, and
- two segments and an arc of a circle intersecting the bisector at point C_2 with coordinates $(e + \epsilon \cos \frac{\alpha}{2}, -\epsilon \sin \frac{\alpha}{2})$.

In the cases of the internal cusp and the turn, the intermediate configuration lies at point C_1 . From convexity arguments, the portion of γ from the origin to c lies under the line (OC_1) and over the x -axis, and therefore inside the tube (figure 10).

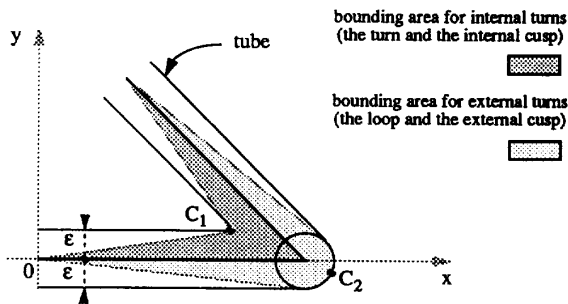


Figure 10: The trajectory γ lies inside the tube of width ϵ .

Similarly, in the cases of the loop and the external cusp, the portion of γ from the origin to c lies over the line (OC_2) and under the x -axis. This suffices to conclude for the case of the external cusp. For the loop, we can check that the curvature of γ at configuration c is less than ϵ . Thus, the first portion lies inside the convex part of the domain bounded by the arc of circle centered at P and of radius ϵ . Therefore, it lies inside the tube. By symmetry with respect to the bisector, the second portion of γ also lies inside the tube. $\square$

Remarks :

1. Note that not only γ is close to the segments in the cartesian space, but also the corresponding path in the configuration space is also as close as we want to the initial path supported by the two segments and the rotation at P : indeed the angular variation of the robot's orientation moving along γ is exactly the same as the initial angular variation along (S_1, P, S_2) . This point is very important in constrained space when collision risks are due to the rotations.
2. In situations geometrically very constrained, the length e of S_1 may be too small and the above computations may lead to $x_0 < 0$. In this case, we fix x_0 to zero. ϵ becomes an indeterminate which can be computed. The final trajectory will then be closer to the segments than the initial geometric constraints of the inputs.

4.7 Comments

All the above computations are founded on the geometric properties of the clothoids and the anti-clothoids. Recall that these curves are fully characterized by the parameters k_c and k_a .

From the kinematic viewpoint, k_c and k_a can be expressed as functions of the velocities v_0 and ω_1 , and the constant acceleration a . If the value of a is not fixed, the trajectory being given, we have one degree of freedom for computing the motion along the trajectory.

5 Conclusion

The presented primitives can be linked together to smooth a reference trajectory composed of oriented broken lines, with constraints on the deviation

from the reference trajectory. Producing an optimal smooth path from the input sequence of segments by the means of an algorithm is still an unsolved problem that we are investigating. The objective is to produce a trajectory like the one shown in figure 11. In this example the cusps and the loop were imposed manually to the system. The swept space during motion is shown in figure 12.

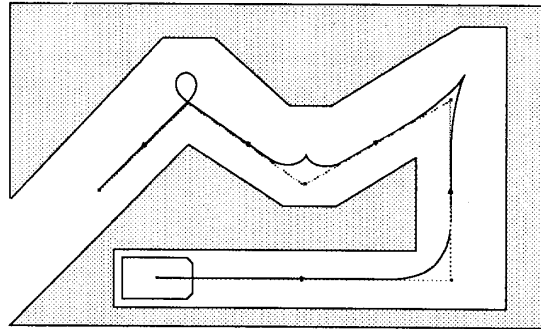


Figure 11: A sequence of primitives linking a turn, an external cusp, an internal cusp, and a loop.

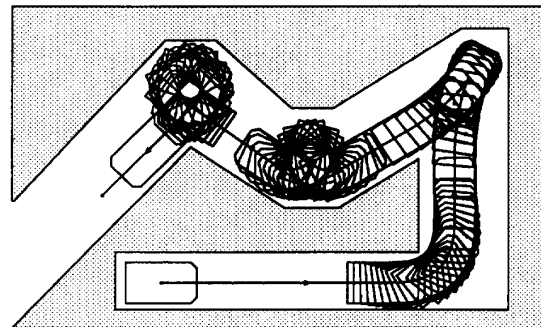


Figure 12: Space swept by the robot during execution of the trajectory of figure 11.

We addressed the case of a sequence Translation-Rotation-Translation. The same approach would have been also possible for the case Rotation-Translation-Rotation. The produced smooth trajectory would pass through the vertices of the reference trajectory.

Finally, we recall that even if the smoothing was produced by "bang-bang" commands, this does not guarantee the optimality of the trajectory. Characterization of the optimal trajectories is still an open problem.

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Appendix: Parametric equations of the anticlothoid

In this appendix, we present the computation of the configuration $c(t)$ of a mobile robot moving along an anticlothoid γ from a start configuration $c_0 = (x_0, y_0, \theta_0)$ with initial linear and angular velocities v_0 and ω_0 .

The coordinates of $c(t)$ are :

$$\begin{cases} x(t) = \int_0^t v(u) \cos \theta(u) du + x_0 \\ y(t) = \int_0^t v(u) \sin \theta(u) du + y_0 \\ \theta(t) = \int_0^t \omega(u) du + \theta_0 \end{cases}$$

From the definition of anticlothoids (see Section 2.3), we have $\omega(t) = \omega_0$ and $v(t) = at + v_0$. Moreover the characteristic parameter of the anticlothoid is $k_a = \frac{a}{\omega_0^2}$. Thus :

$$\begin{cases} x(t) = \int_0^t (au + v_0) \cos(\omega_0 u + \theta_0) du + x_0 \\ y(t) = \int_0^t (au + v_0) \sin(\omega_0 u + \theta_0) du + y_0 \\ \theta(t) = \omega_0 t + \theta_0 \end{cases}$$

By using an integration by parts, the coordinates $x(t)$ and $y(t)$ as functions of $\theta(t)$ are :

$$\begin{aligned} x(t) &= k_a (\cos \theta - \cos \theta_0 + (\theta - \theta_0 + \frac{\omega_0 v_0}{a}) \sin \theta - \frac{\omega_0 v_0}{a} \sin \theta_0) + x_0 \\ y(t) &= k_a (\sin \theta - \sin \theta_0 - (\theta - \theta_0 + \frac{\omega_0 v_0}{a}) \cos \theta + \frac{\omega_0 v_0}{a} \cos \theta_0) + y_0 \\ \theta(t) &= \omega_0 t + \theta_0 \end{aligned}$$

When $v_0 < 0$, there is a point $\gamma(t)$ such that $v(t) = 0$. This is a cusp. The corresponding configuration $\tilde{c}$ verifies :

$$\begin{aligned} x_{\tilde{c}} &= k_a (\cos \theta_{\tilde{c}} - \cos \theta_0 + (\theta_{\tilde{c}} - \theta_0) \sin \theta_0) + x_0 \quad (11) \\ y_{\tilde{c}} &= k_a (\sin \theta_{\tilde{c}} - \sin \theta_0 - (\theta_{\tilde{c}} - \theta_0) \cos \theta_0) + y_0 \quad (12) \\ \theta_{\tilde{c}} &= -\frac{v_0 \omega_0}{a} + \theta_0 \quad (13) \end{aligned}$$

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